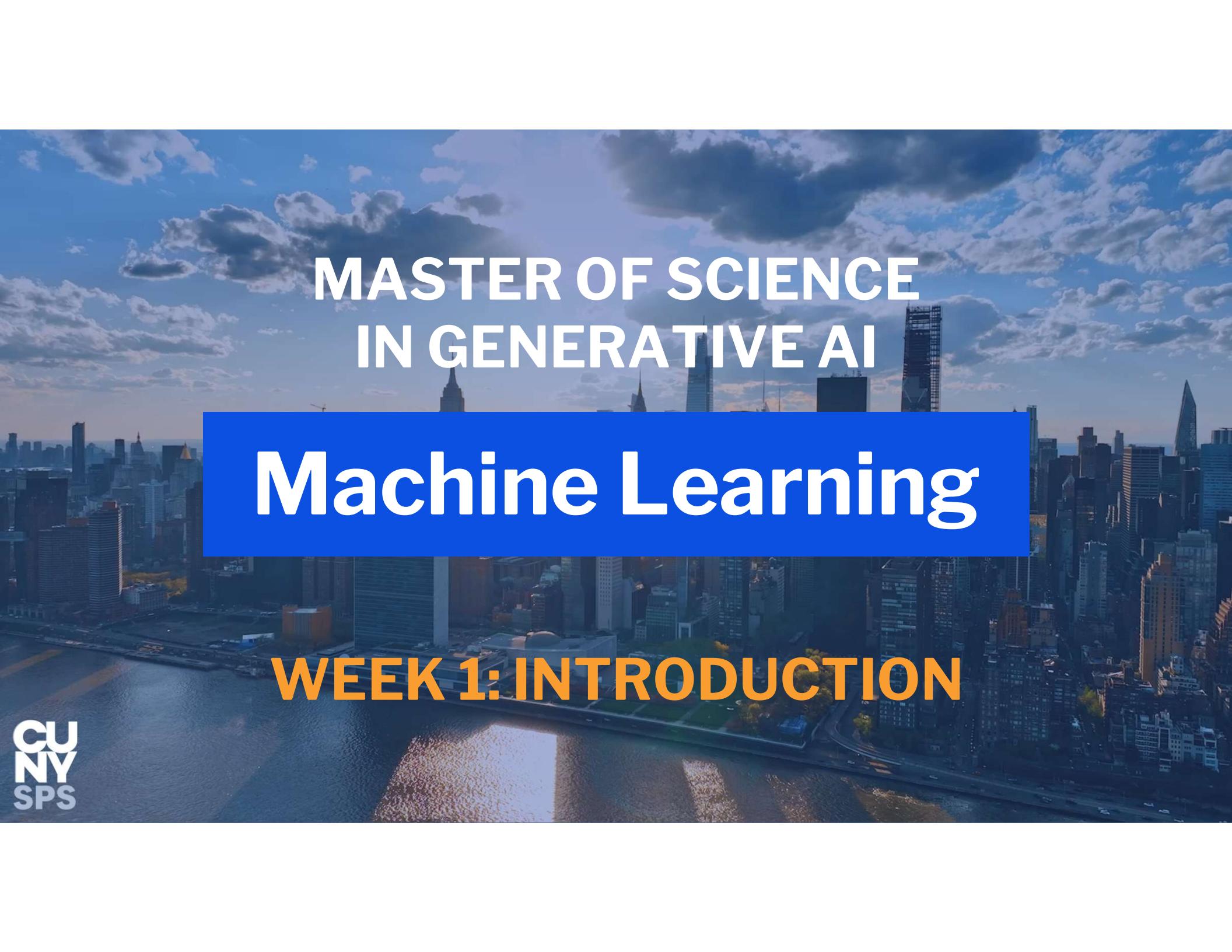




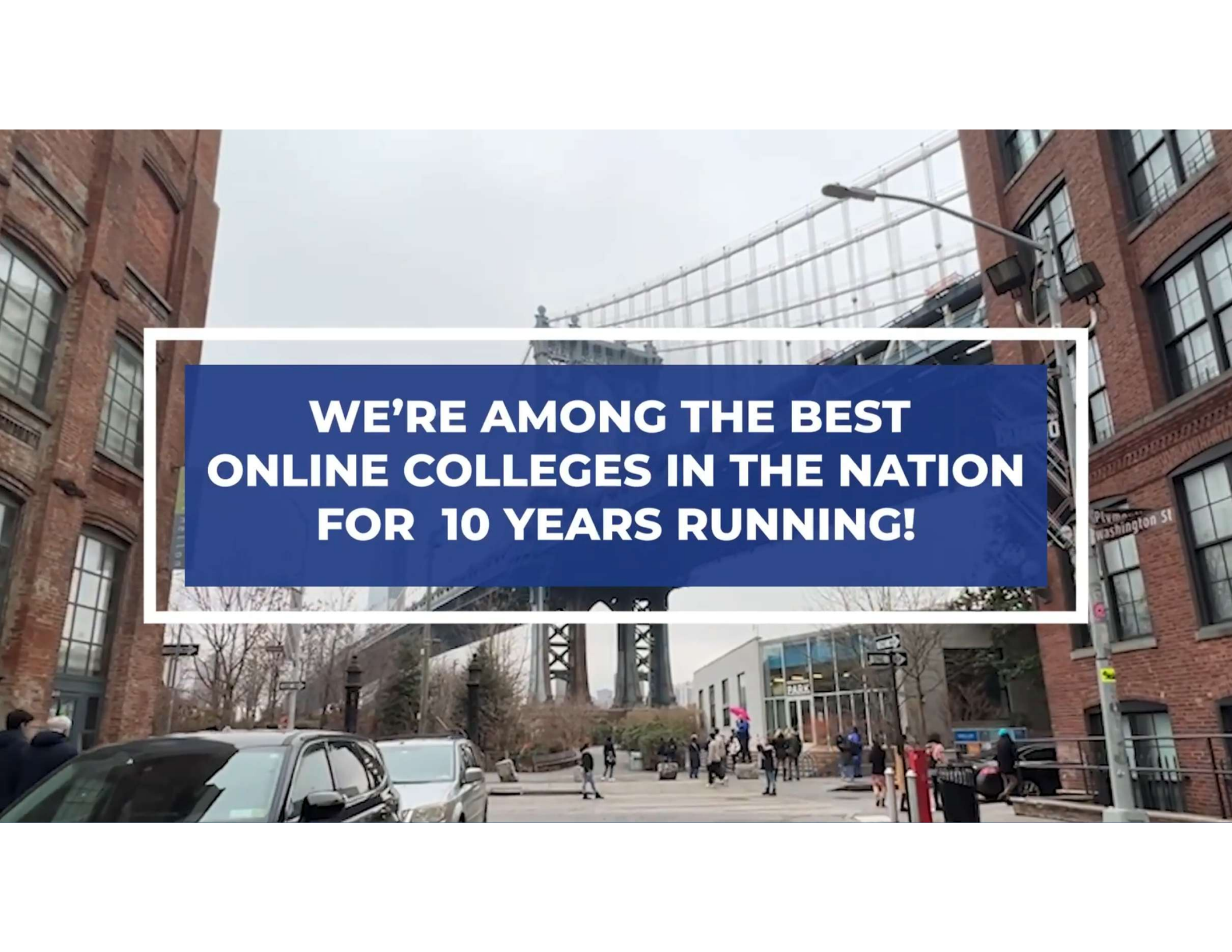
**MASTER OF SCIENCE
IN GENERATIVE AI**

Machine Learning

WEEK 1: INTRODUCTION

A photograph of a busy New York City street corner. In the foreground, a cyclist wearing a bright neon green jacket and a black helmet is riding a bicycle towards the camera. The street is filled with yellow taxis and other vehicles. In the background, there are tall buildings with many windows. A large blue rectangular box with white text is overlaid in the center of the image. The text reads:

**HIGHEST RANKED
ONLINE COLLEGE IN
NEW YORK CITY!**

A street scene in Pittsburgh, featuring brick buildings on either side and the iconic steel structure of the Fort Pitt Bridge in the background. A large blue rectangular overlay with white text is centered in the upper half of the image. In the foreground, there are parked cars and pedestrians on the sidewalk. A street sign for 'Plymouth Washington St' is visible on the right building.

**WE'RE AMONG THE BEST
ONLINE COLLEGES IN THE NATION
FOR 10 YEARS RUNNING!**



Top Data Science Programs Integrating AI, ML, Data Engineering, and Generative AI

Leading universities are integrating AI, machine learning, data engineering, MLOps, and generative AI into data science programs, preparing professionals for end-to-end AI development and enterprise deployment.

Written By : Poulami Saha

Reviewed By : Achu Krishnan

Published: 24th Jun, 2026 at 11:45 PM

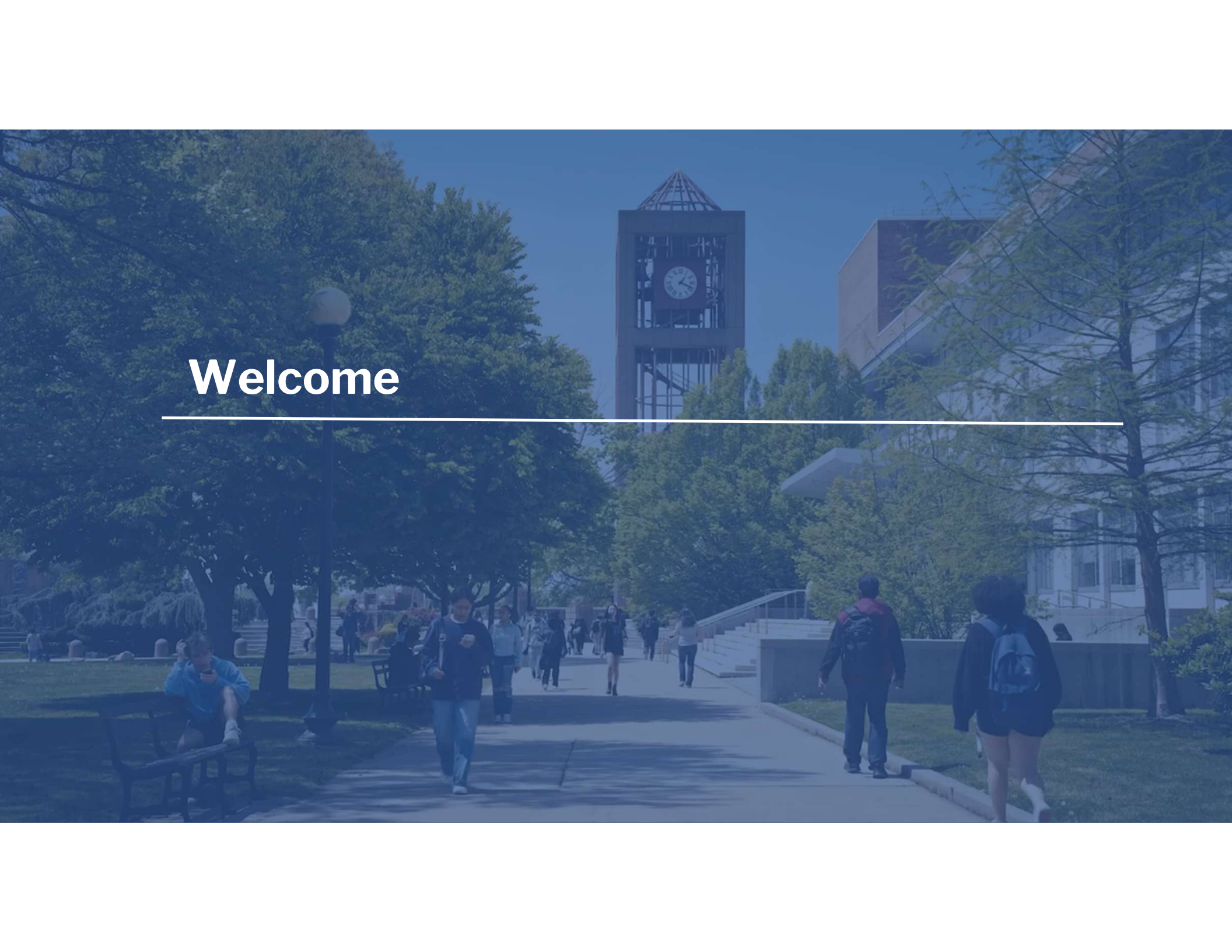
Top 7 Data Science Programs Integrating AI, ML, Data Engineering and Generative AI

Institution	AI & ML	Data Engineering	Generative AI	Industry Focus	Best For
MIT	Excellent	Strong	Strong	Research & Innovation	AI Researchers
CMU	Excellent	Strong	Strong	Applied AI	ML Engineers
Stanford	Excellent	Strong	Strong	Startups & Innovation	AI Entrepreneurs
UC Berkeley	Strong	Excellent	Strong	Enterprise Analytics	Data Scientists
TU Delft	Strong	Strong	Moderate	Responsible AI	AI System Designers
IIT Jammu	Strong	Strong	Excellent	Enterprise AI	Working Professionals
CUNY SPS	Moderate	Moderate	Excellent	Generative AI	GenAI Specialists



Introduction to Machine Learning (GAI601)

Joe Sabelja



Welcome

Syllabus

1. Introduction to Machine Learning
2. Linear and Logistic Regression Models
3. Classification: Discriminant Analysis & k-Nearest Neighbors (kNN)
4. Classification: Naive Bayes & Integrated Methods
5. Decision Trees
6. Ensemble Methods: Bagging, Random Forests, and Boosting (Part 1)
7. Ensemble Methods: Boosting (Part 2) and Additive Trees
8. Support Vector Machines (SVMs)
9. Unsupervised Learning: Clustering
10. Unsupervised Learning: PCA and Distance Metrics
11. Resampling, Model Selection, and Additional Topics
12. Bias-Variance Tradeoff
13. Neural Networks and Foundation Models
14. Ethics and Responsible AI/ML

Schedule

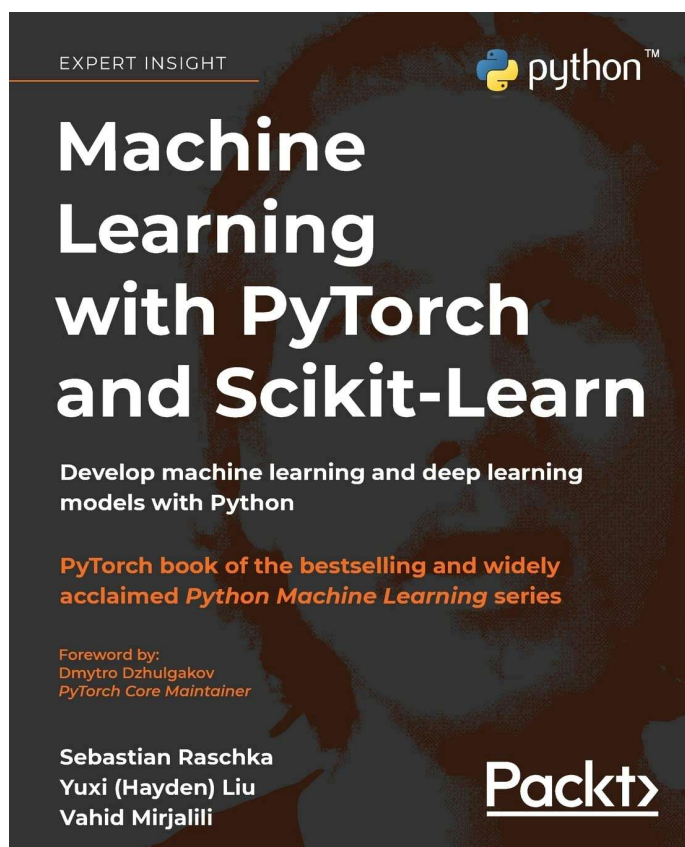
2026	September					
MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
31 Week 1	01	02	03	04	05	06
07 Labor Day	08 Week 2	09	10	11 Rosh Hashana	12	13
14 Week 3	15	16	17	18	19	20 Yom Kippur
21 Yom Kippur	22 Week 4	23	24	25	26	27
28 Week 5	29	30	01	02	03	04

2026	October					
MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
28	29	30	01	02	03	04
05 Week 6	06	07	08	09	10	11
12 Columbus Day	13 Week 7	14	15	16	17	18
19 Week 8	20	21	22	23	24	25
26 Week 9	27	28	29	30	31	01

2026	November					
MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
26	27	28	29	30	31	01
02 Week 10	03	04	05	06	07	08
09 Week 11	10	11	12	13	14	15
16 Week 12	17	18	19	20	21	22
23	24	25	26	27	28	29
Thanksgiving: CUNY SPS offices closed, no classes scheduled*						

2026	December					
MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
30 Week 13	01	02	03	04	05	06
07 Week 14	08	09	10	11	12	13
14	15	16	17	18	19	20
21 Fall Semester Ends	22	23	24 Holiday observed - CUNY SPS is closed	25 Final Grades are due	26	27
28	29	30	31 Holiday observed - CUNY SPS is closed	01	02	03

Textbook



Machine Learning with PyTorch and Scikit-Learn

by Sebastian Raschka, Yuxi (Hayden) Liu, Vahid Mirjalil

February 2022

ISBN-10: 1801819319

ISBN-13: 978-1801819312

This is not a programming course

To help you understand the technical concepts and what data scientists do we will do vibe coding.

If you prefer R let me know and I will provide alternatives

Class Policies

1. Please leave your camera on

- See: https://www.cuny.edu/wp-content/uploads/sites/4/page-assets/academics/faculty-affairs/Camera-Use-Guidance-for-Online-and-Hybrid-Courses_FINAL-JUNE-20-2024.pdf

2. Review the Course Policies on Brightspace

- Respect, Online Etiquette and Anti-Harassment
- Lateness Policy

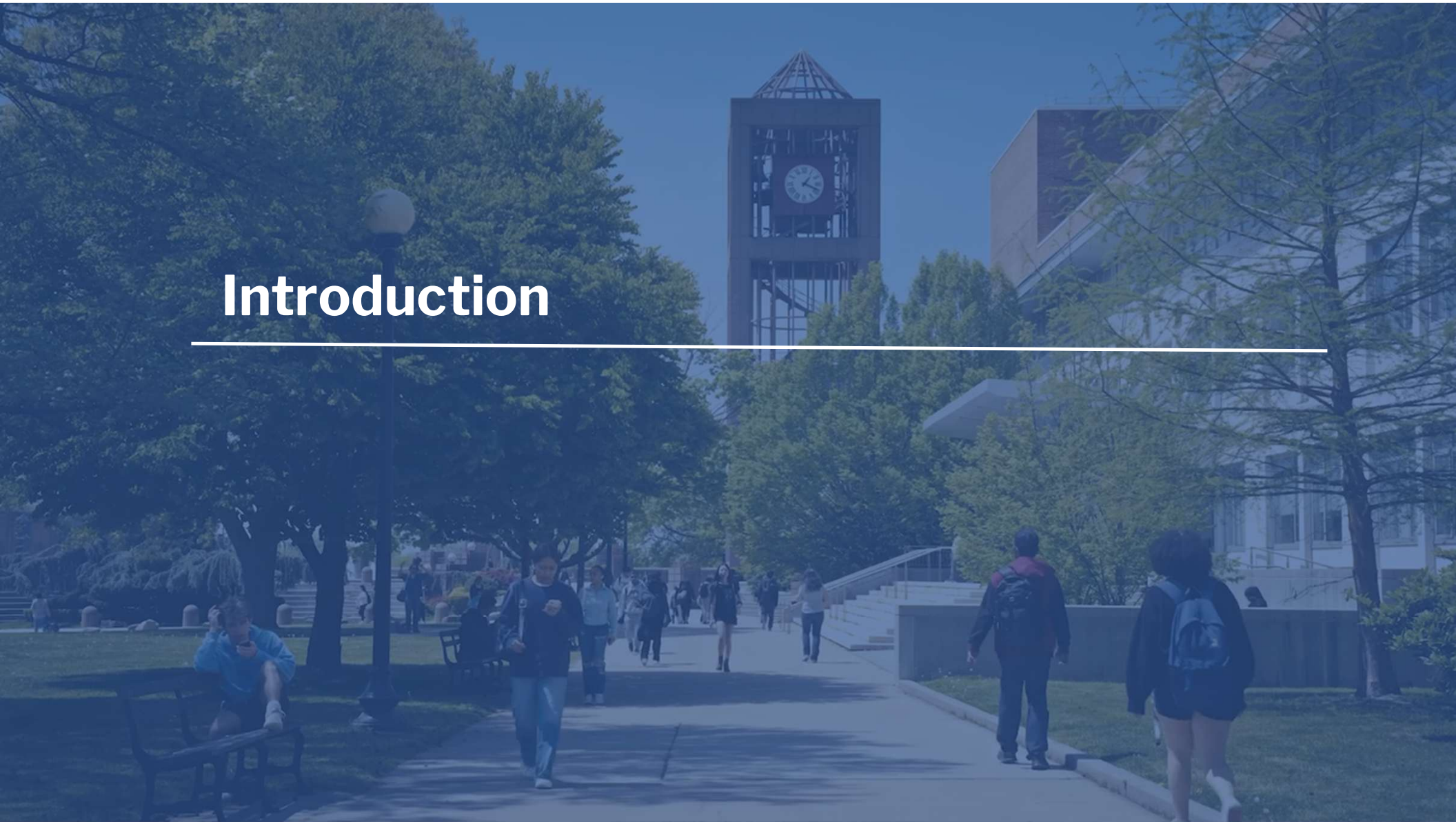
3. Read the guidelines on Generative AI & Academic Integrity on Brightspace

- Provide Snapshots with answers
- Do not use Generative AI for quizzes and exams
- You (the human) manage AI, not the other way around
- Don't cheat yourself

Grading

Component	Number	Total Value
Quizzes	5	20%
Assignments	4	20%
Exam: Mid-term	1	30%
Exam: End of semester	1	30%
		100%

Introduction

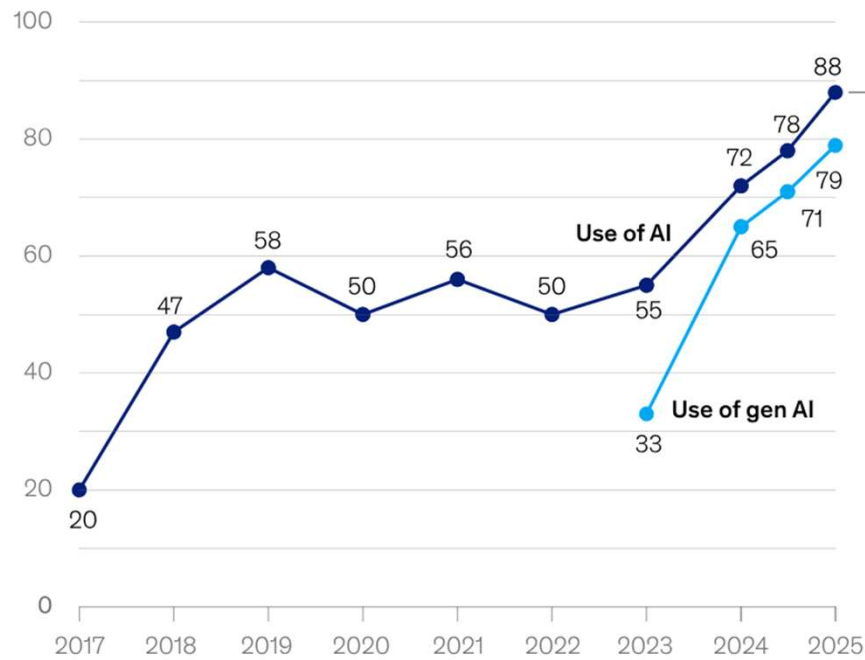


AI has become a critical need for firms

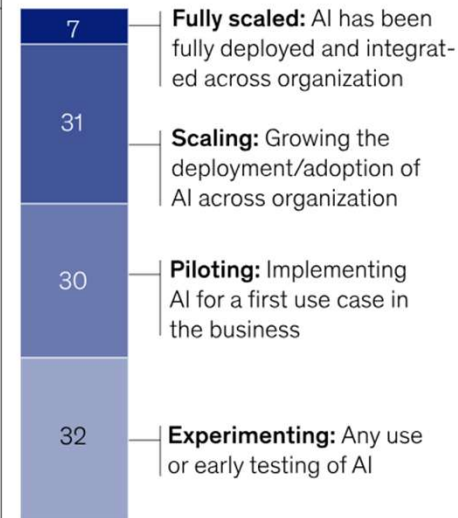
Reported use of AI in at least one business function continues to increase.

Use of AI by respondents' organizations, % of respondents

Organizations that use AI in at least 1 business function¹



Phase of AI use among organizations using AI in 2025

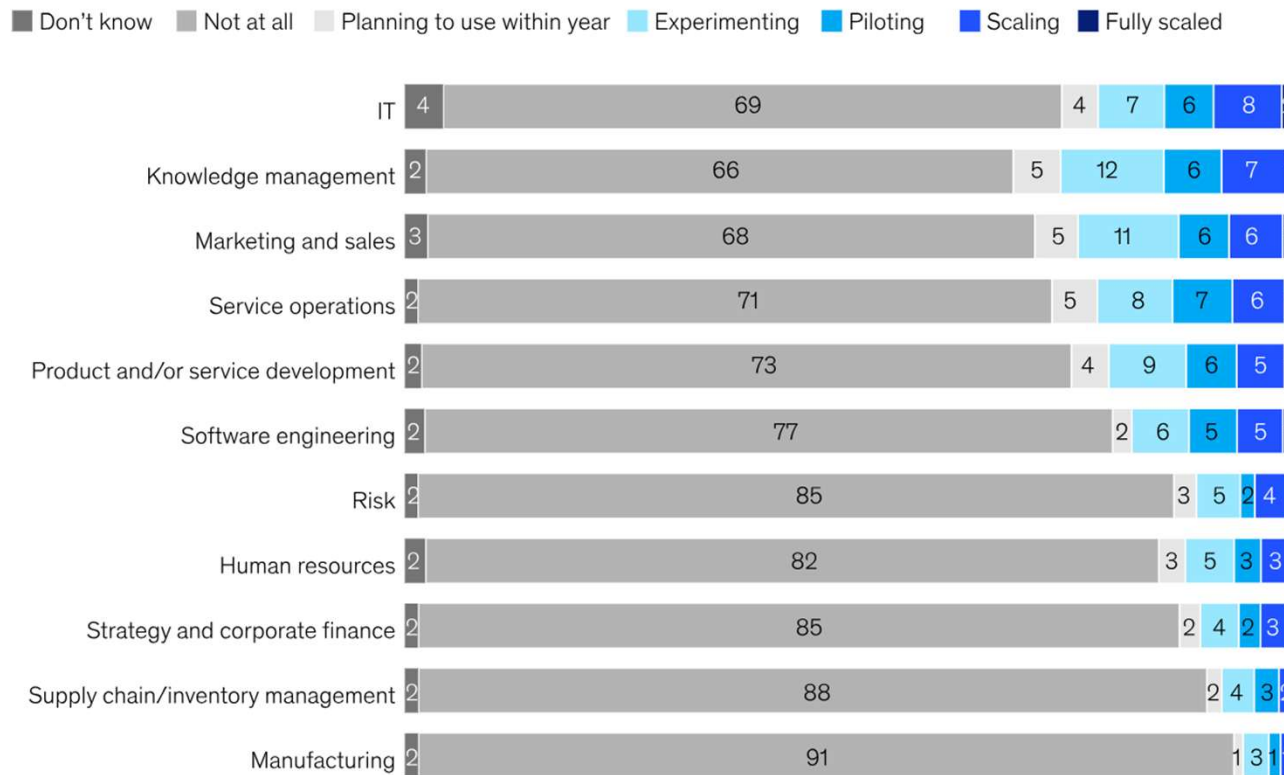


Source: McKinsey

Real adoption of Generative AI is slow

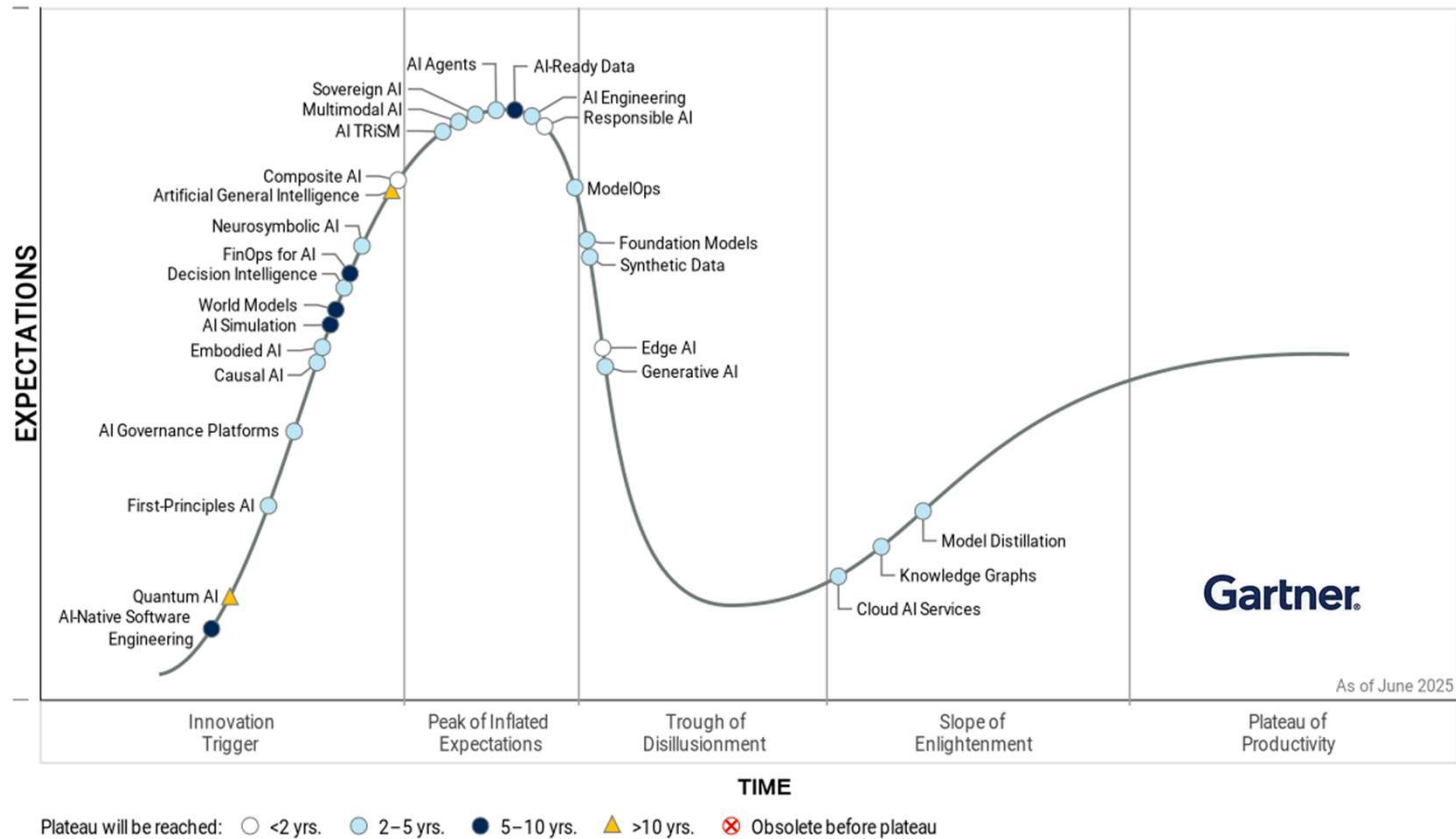
No more than 10 percent of respondents report scaling AI agents in any individual function.

Phase of AI agent use at respondents' organizations, by business function,¹ % of respondents (n = 1,933)



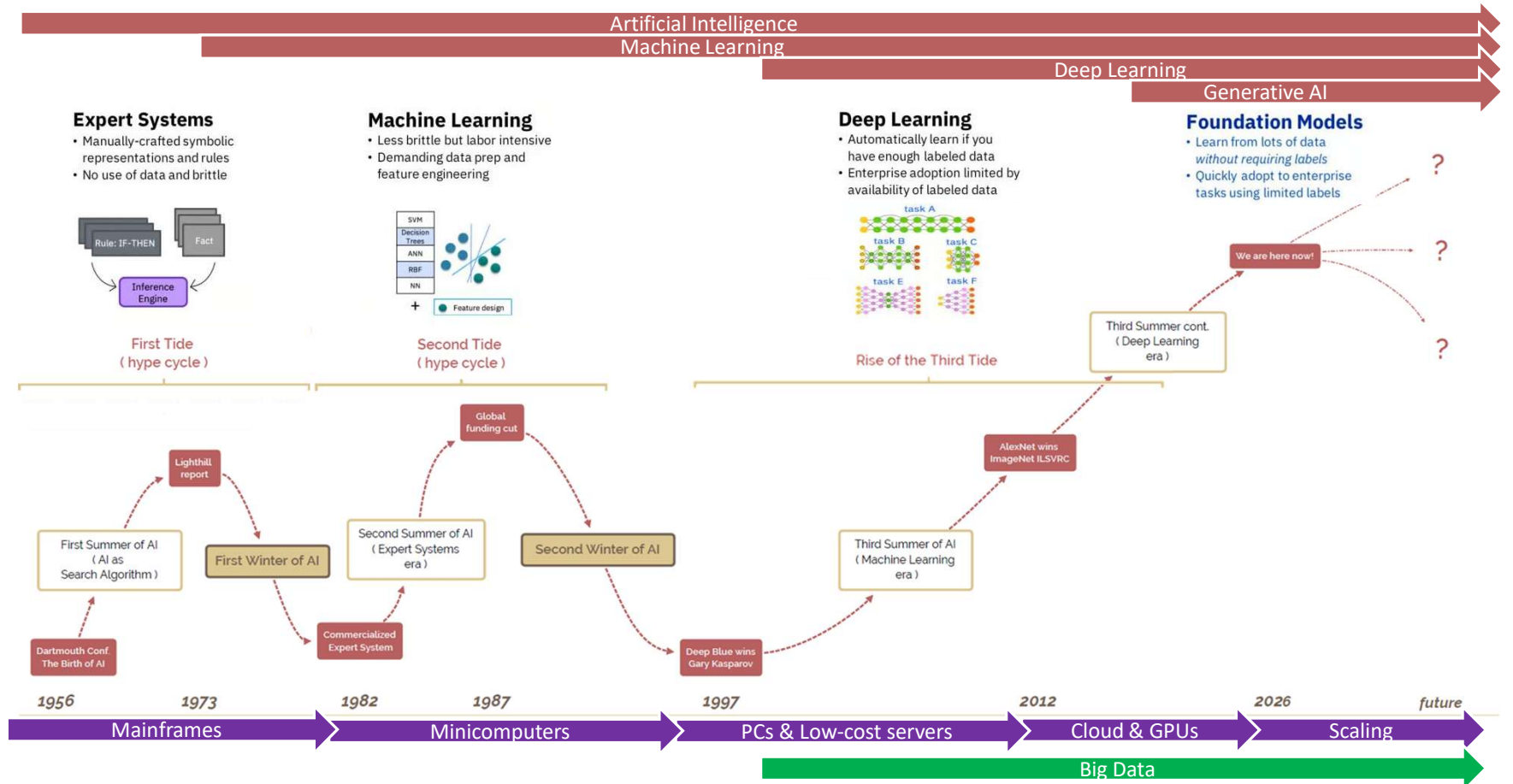
Source: McKinsey

There is a lot of excitement



Source: Gartner

AI Winters



Source: A brief history of AI, Amirhosein Toosi, et al, 2022

[illegible]

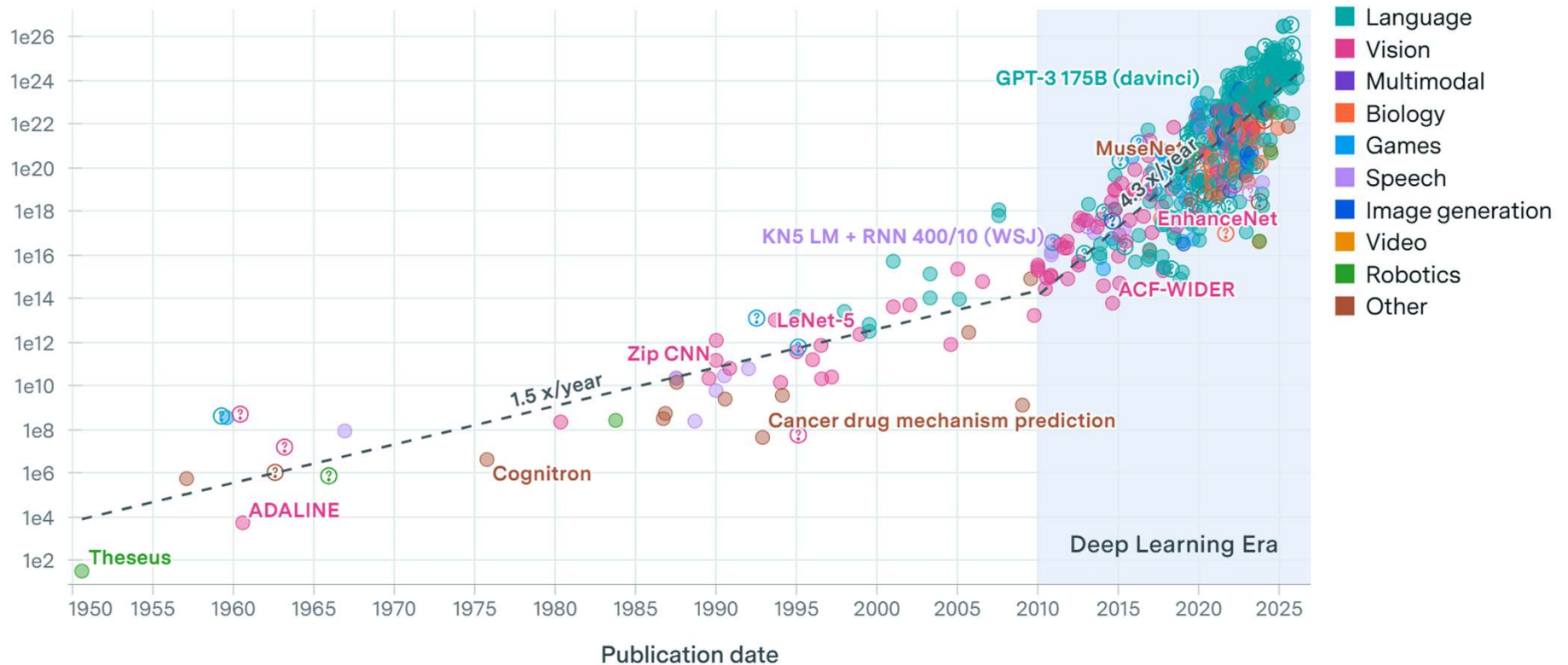
Growth is accelerating – due to GPUs

Notable AI models

EPOCH AI

Training compute (FLOP)

② : Speculative data 540 Results Domain



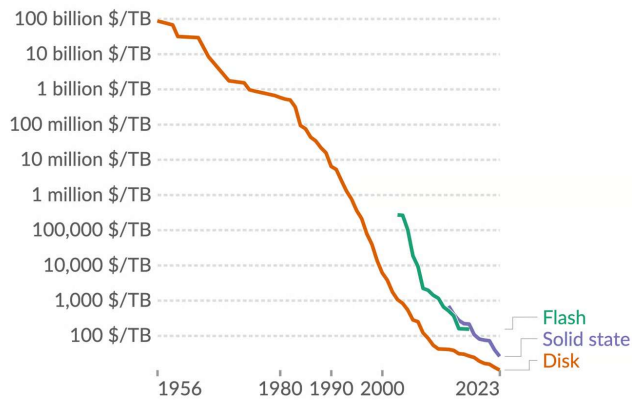
Source: epoch.ai

AI is becoming cheaper

Historical price of computer storage

Our World in Data

Expressed in US dollars per terabyte (TB), adjusted for inflation. "Disk" refers to magnetic storage, "flash" to memory used for rapid data access and rewriting, and "solid state" to solid-state drives (SSDs).



Data source: John C. McCallum (2023)

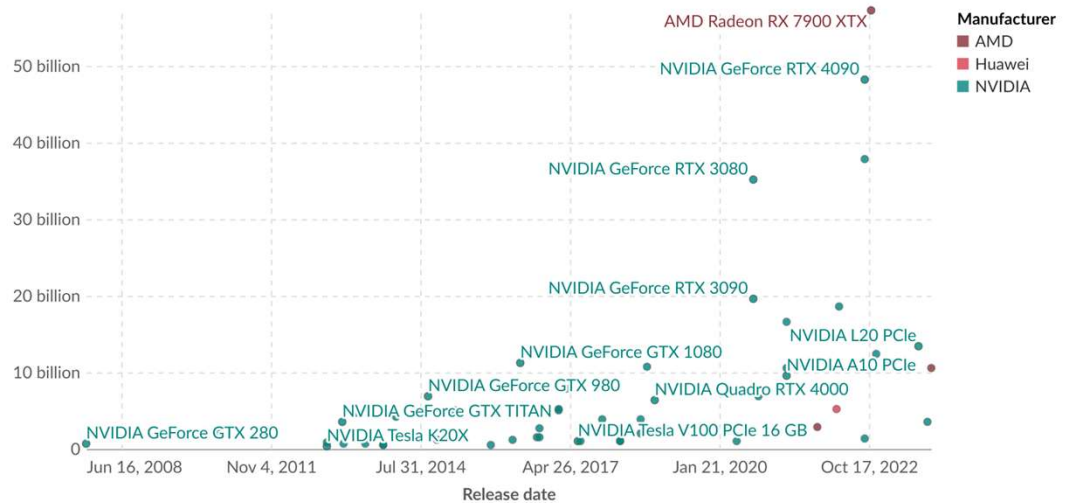
CC BY

GPU computational performance per dollar

Our World in Data

Hardware computational performance shown in floating-point operations¹ per second (FLOP/s) per US dollar, adjusted for inflation.

GPU performance per dollar (FLOP/s/\$)



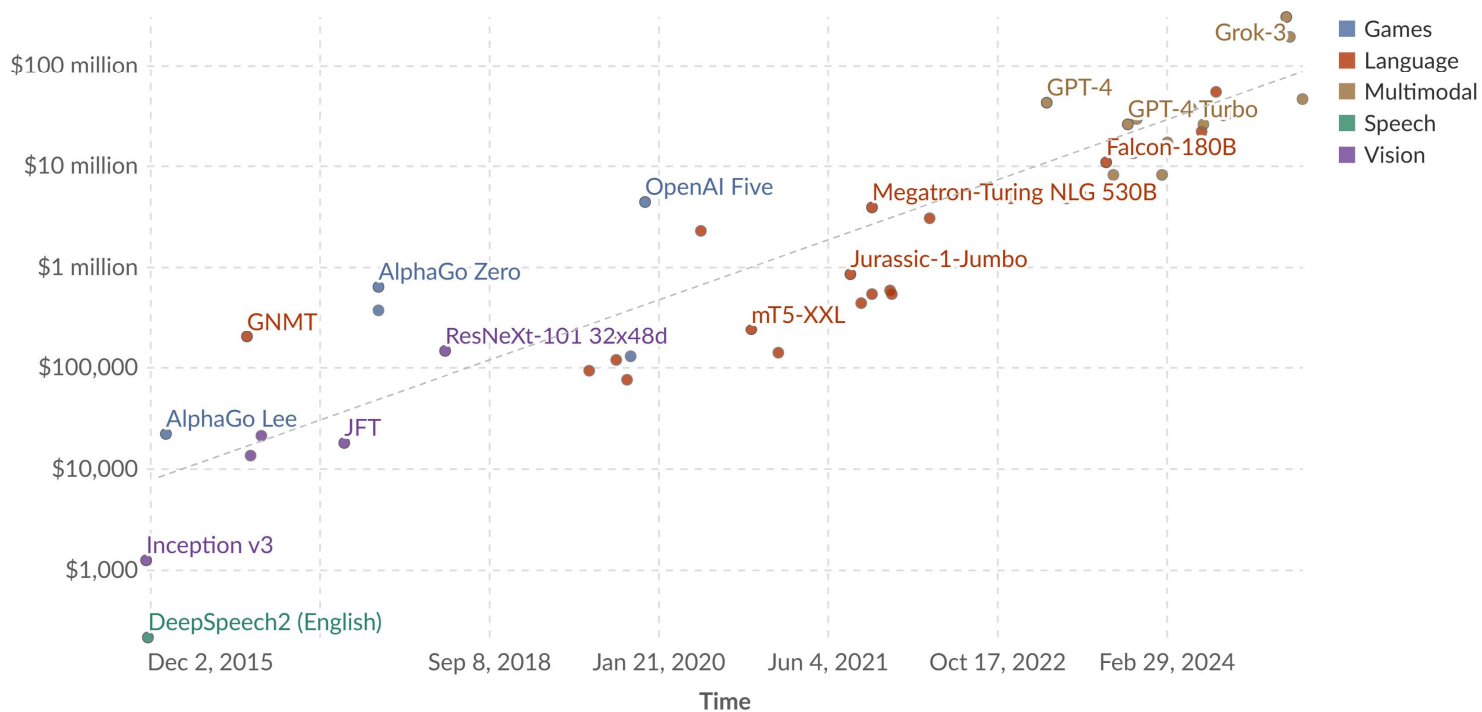
Source: Our world in data

Generative AI models are getting bigger

Hardware and energy cost to train notable AI systems

This data is expressed in US dollars, adjusted for inflation.

Cost (constant 2023 US\$; plotted on a logarithmic axis)

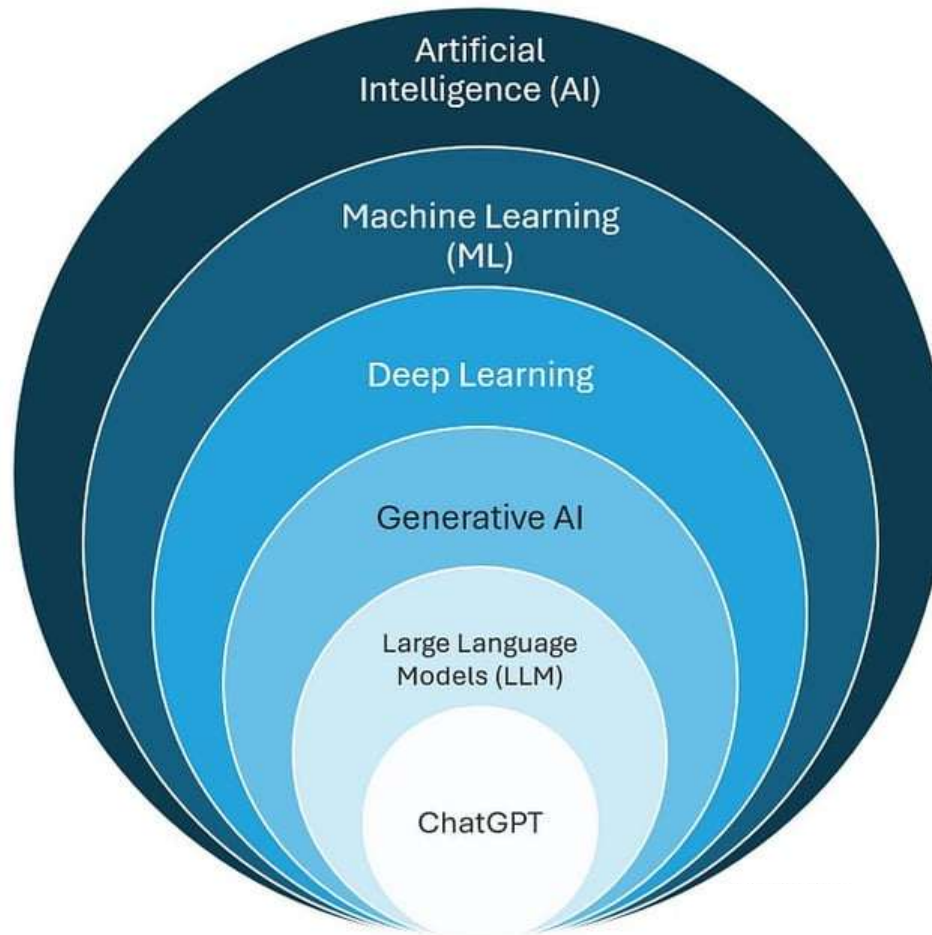


Source: Our world in data

A blue-tinted photograph of a university campus. In the background, a tall clock tower with a glass-enclosed top and a visible clock face stands prominently. To its right is a modern, multi-story building with a grid-like facade. The foreground is filled with lush green trees and a wide, paved pedestrian walkway. Several students are walking along the path; some are carrying backpacks, and one student is sitting on a bench on the left. The overall atmosphere is that of a busy academic environment.

What is Artificial Intelligence?

What is Artificial Intelligence?



What is intelligence?

"A fundamental problem in artificial intelligence is that nobody really knows what intelligence is."

-Legg & Hutter (2007)

"... the ability to solve hard problems." M. Minsky

"The ability to use memory, knowledge, experience, understanding, reasoning, imagination and judgement in order to solve problems and adapt to new situations." AllWords Dictionary, 2006

"The capacity to acquire and apply knowledge." The American Heritage Dictionary, fourth edition, 2000

"Intelligence is not a single, unitary ability, but rather a composite of several functions. The term denotes that combination of abilities required for survival and advancement within a particular culture." A. Anastasi

"Intelligence is the ability for an information processing system to adapt to its environment with insufficient knowledge and resources." P. Wang

"... the ability to plan and structure one's behavior with an end in view." J. P. Das

"... doing well at a broad range of tasks is an empirical definition of 'intelligence' " H. Masum

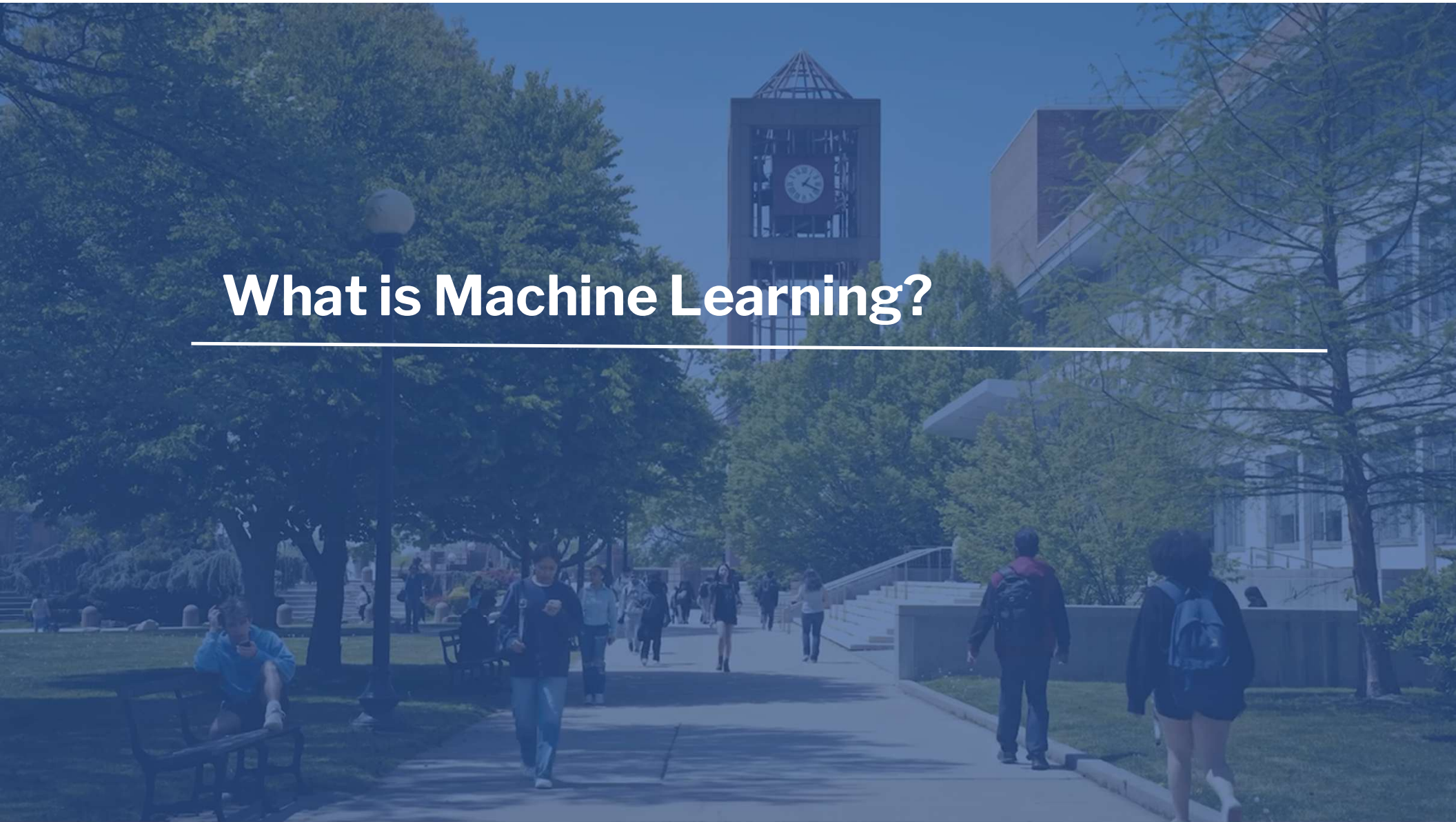
"... the ability of a system to act appropriately in an uncertain environment, where appropriate action is that which increases the probability of success, and success is the achievement of behavioral subgoals that support the system's ultimate goal." J. S. Albu

"Intelligence measures an agent's ability to achieve goals in a wide range of environments."

S. Legg and M. Hutter

Source: A Collection of Definitions of Intelligence, Shane Legg, Marcus Hutter, 2007 arXiv:0706.3639

What is Machine Learning?



What is Data Science?

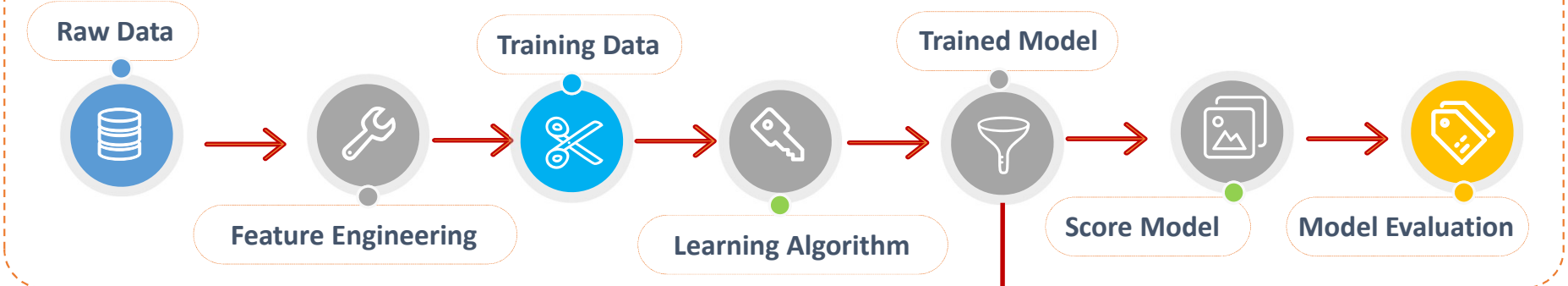
“At a high level, data science is a set of fundamental principles that guide the extraction of knowledge from data.”

Principles can be statistical, computational, algorithmic, visual, etc.

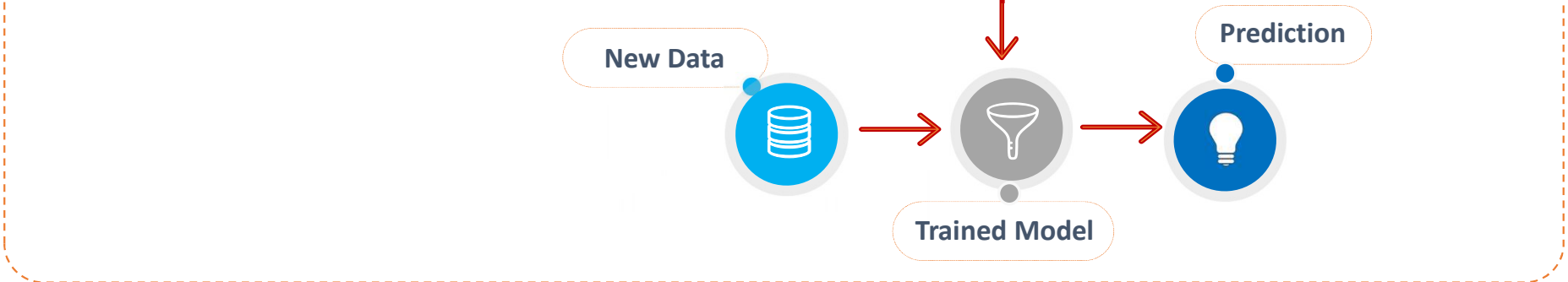
Source: NVIDIA DLI Accelerated Data Science

How does Machine Learning work?

TRAINING



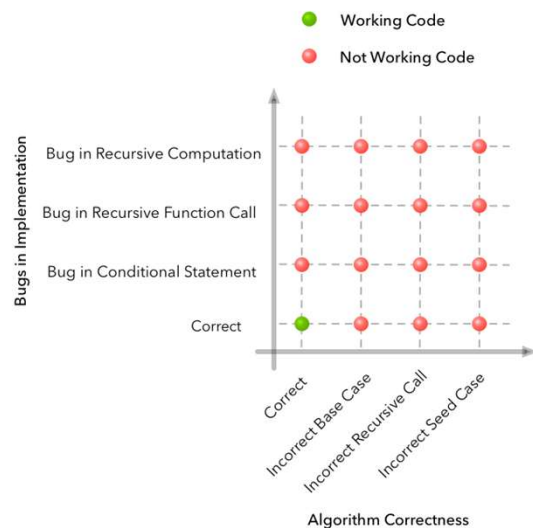
INFERENCE



Why is Data Science hard?

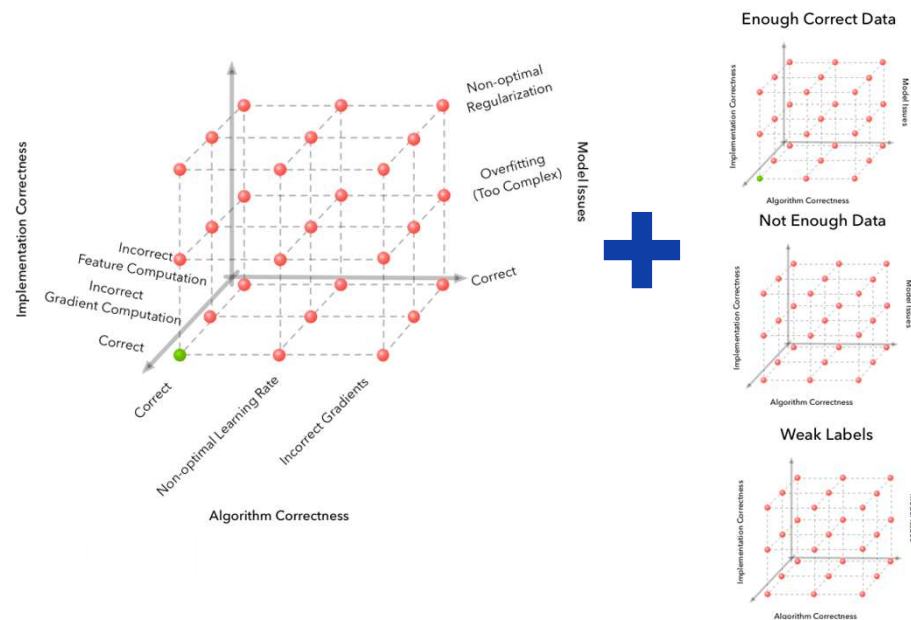
Software Development is a Two-dimensional Problem

Algorithm selection and Software implementation



Machine Learning is a Four-dimensional Problem

Software development + Model Implementation + Data



Source: <https://ai.stanford.edu/~zayd/why-is-machine-learning-hard.html>

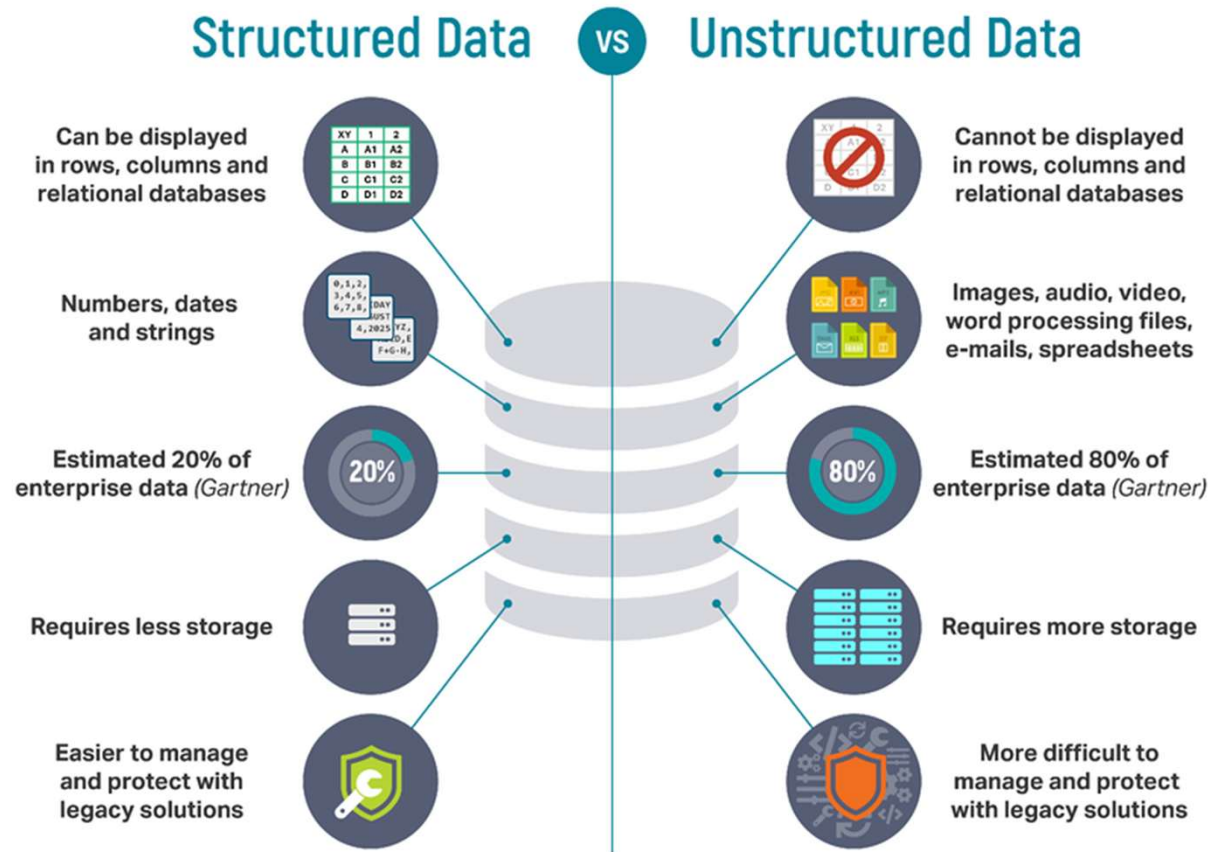
Why is Data Science complex?

- 1. Data science is interdisciplinary**
- 2. Applicable to many domains (e.g., sciences, finance, healthcare, etc.)**
- 3. Big Data**
- 4. Analysis of data**
- 5. Statistics (a pillar of data science)**
- 6. Experimentation**
- 7. Constant Evolution**
- 8. Data Wrangling**

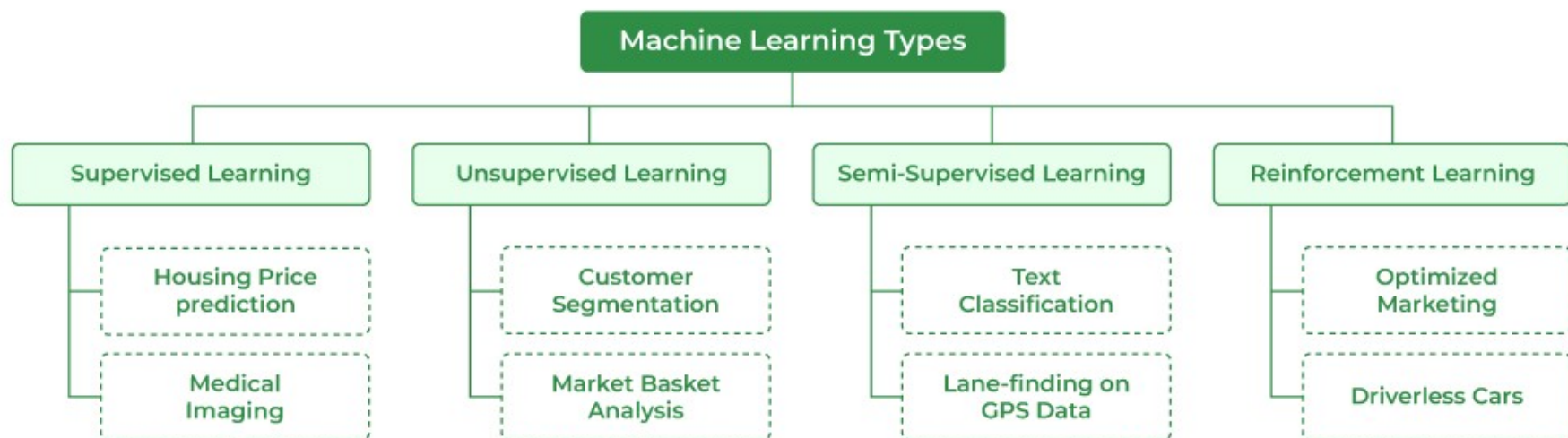
A blue-tinted photograph of a university campus. In the background, a tall clock tower with a glass-enclosed clock face and a pyramidal roof stands prominently. To its right is a modern, multi-story building with a grid-like facade. The foreground is filled with lush green trees and a wide, paved pedestrian walkway. Several students are visible: some are walking along the path, while one student is sitting on a wooden bench on the left. A black lamppost with a white globe is positioned on the left side of the path. The overall atmosphere is calm and academic.

Types of Machine Learning

Types of data



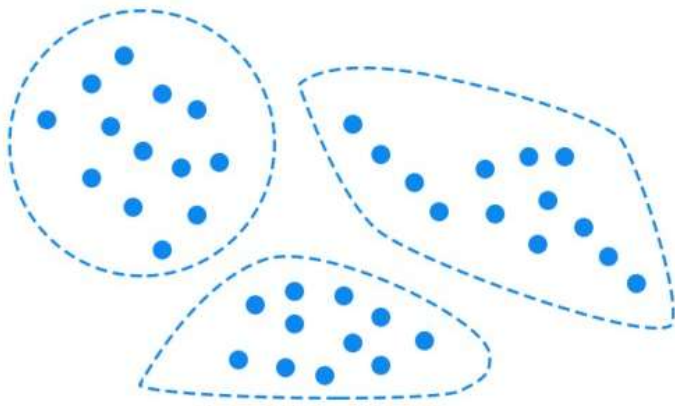
Types of Machine Learning



Supervised vs Unsupervised Learning

UNSUPERVISED

Clustering

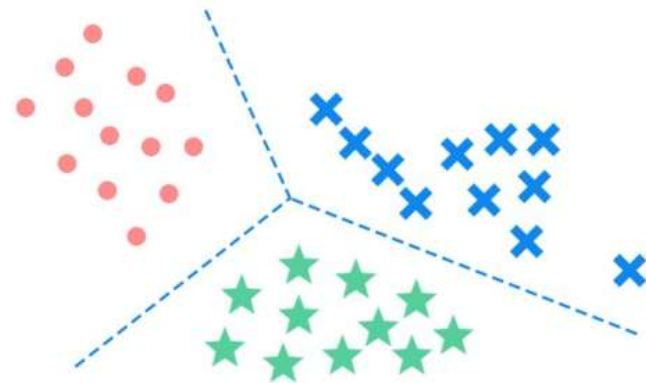


Business use-cases:

- Customer segmentation
- Cross- selling
- Anomaly detection e.g. Expenses
- Document clustering
- Recommendation & personalization
- Data Reporting (dimensionality Reduction)

SUPERVISED

Classification

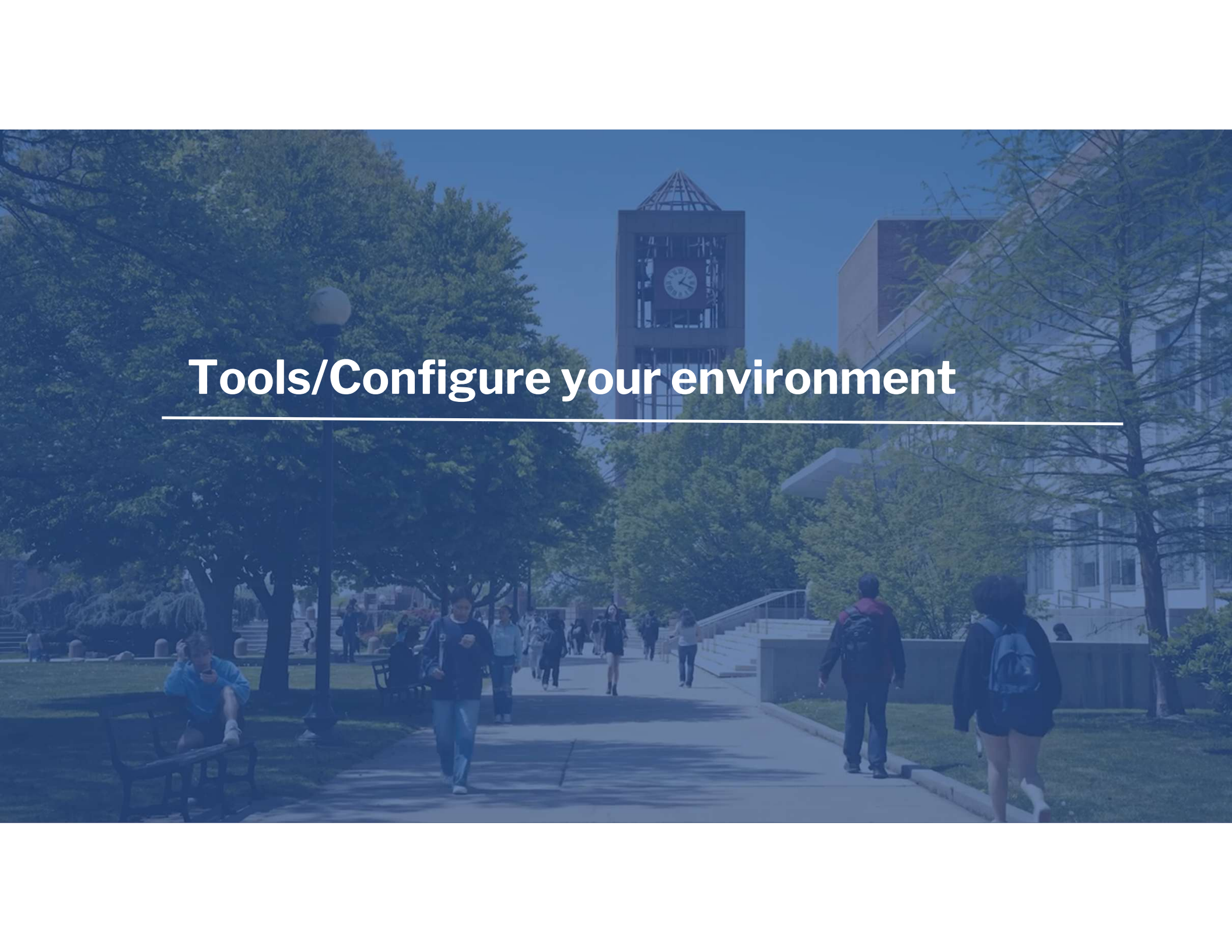


Business use-cases:

- Loan approval, credit risk assessment
- Credit Card Fraud Detection
- Sentiment analysis, text classification
- Speech recognition, Image classification
- Spam Detection
- Stock Price Prediction

Types of Algorithms



A blue-tinted photograph of a university campus. In the background, a tall clock tower with a glass-enclosed clock face and a pyramidal roof stands prominently. To its right is a modern, multi-story building with a grid-like facade. The foreground shows a wide, paved pedestrian walkway lined with mature trees. Several students are walking along the path; some are carrying backpacks. On the left, a student is sitting on a wooden bench. The overall atmosphere is that of a busy academic environment.

Tools/Configure your environment

Register for free tools

1. Colab

- Go to <https://colab.research.google.com/signup>
- Create a Google account (if you don't have one)
- Sign up for **Colab Pro for Education** (free for students)

2. GitHub Copilot

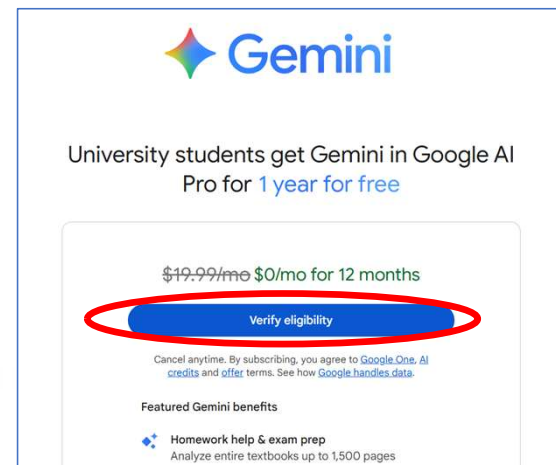
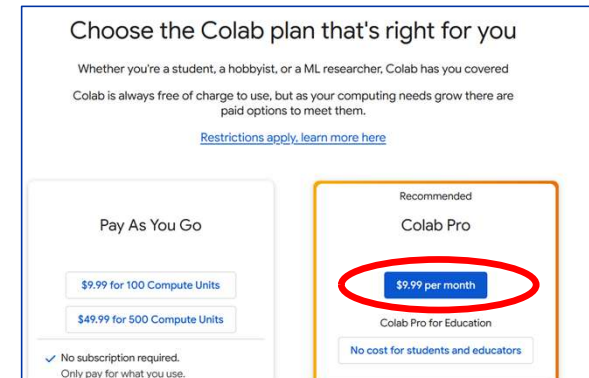
- Go to <https://github.com/education>
- Click on **Join GitHub Education**
- Create a GitHub account (if you don't have one)
- Click on Start an Application to get access

3. Google Gemini

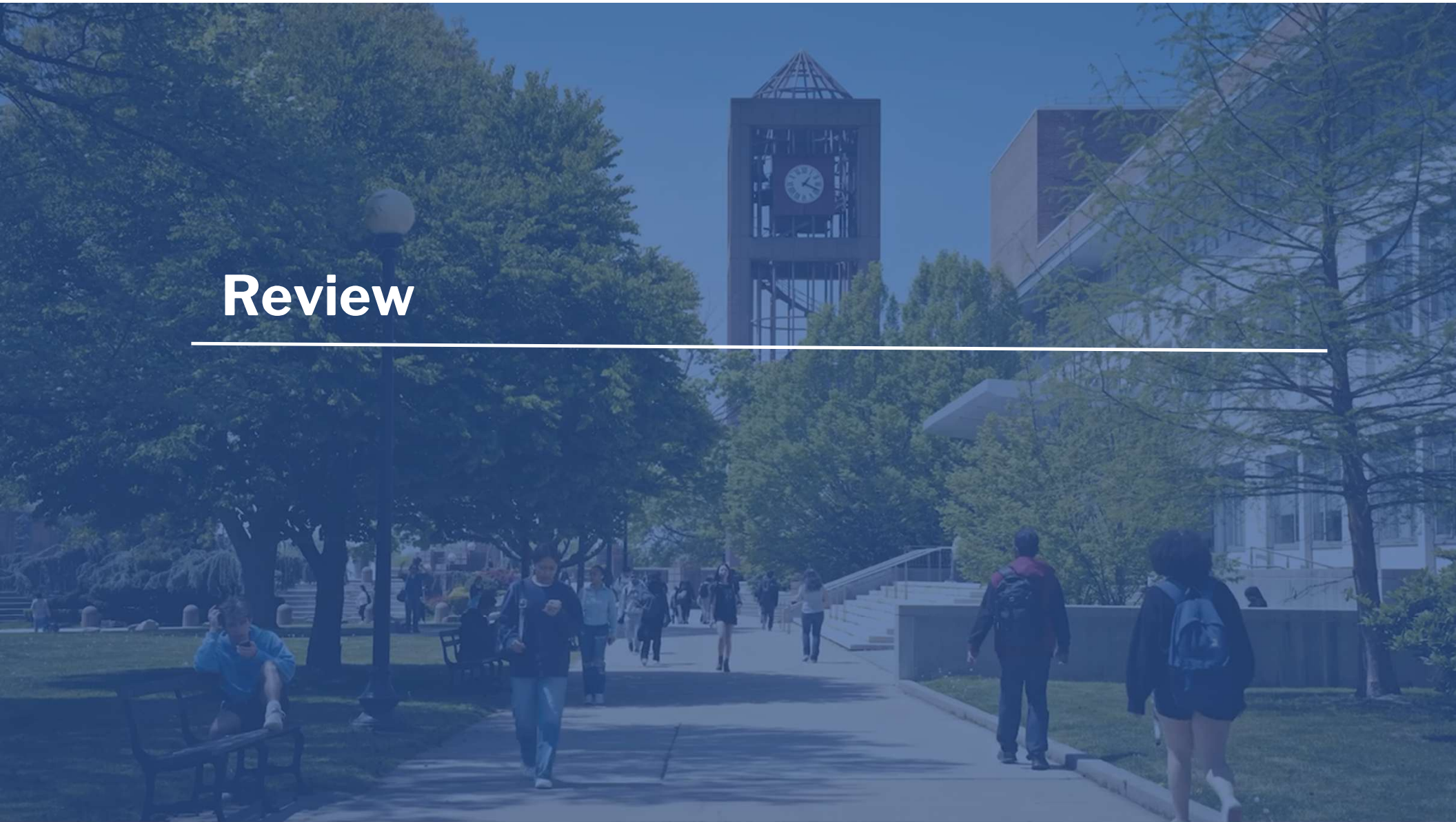
- Go to <https://gemini.google/us/students>
- Fill out the verification form

4. Microsoft Copilot 365

- Free with you Office 365 access (via CUNY)



Review



This week we covered

1. Supervised & Unsupervised Learning

Define the basic types of machine learning (supervised, unsupervised) using examples from real-world applications.

2. Business Role

Explain the role of machine learning in modern business contexts and decision-making.

3. Tooling

Set up and configure Python (or R) environments for machine learning tasks.

4. ML Algorithms

Identify appropriate ML techniques for structured and unstructured business data.

An aerial photograph of the Manhattan skyline, featuring the Manhattan Bridge in the foreground and the dense cityscape in the background. The bridge's stone towers and suspension cables are prominent. The water of the Hudson River is visible in the foreground. The sky is clear and blue.

CU NY | School of Professional Studies